

HFT3 VectorBT -> Robustness Bridge Findings

Developer-ready diagnostic and remediation prompt for the VectorBT handoff choke point

Purpose

This document consolidates the current findings around the VectorBT screening run, the zero-candidate HftBacktest handoff, and the suspected robustness bridge defect. It is written as a developer-facing implementation prompt. The goal is to avoid relaunching expensive compute blindly and instead validate whether the bridge is mathematically choking viable candidates before HftBacktest realism replay.

Core conclusion: do not restart the full pipeline yet. Run a robustness bridge sensitivity diagnostic from existing VectorBT artifacts first.

1. Situation Summary

The observed outcome is not that VectorBT found no opportunity. VectorBT promoted candidates. The choke point is the next bridge between screening artifacts and robustness/HftBacktest packaging.

Item	Observed/Interpreted Result
VectorBT promoted candidates	1,728 candidates promoted
Robustness packaging accepted	0 / 50 model families
HftBacktest eligible candidates	0
Likely choke point	scripts/build_robustness_raw_inputs_from_screening.py
Suspicious logic	train_event = events[0] used for family-level surface stability
Action	Run bridge diagnostic before any full rerun

2. Methodology Diagnosis

The suspicious line is not merely a code smell; it is a quant methodology problem. If a family-level gate computes surface stability from only the first event in the family, then one arbitrary event can reject an entire model family before walk-forward, fold-level, WFC, DSR, PBO, or CSCV evidence is even constructed.

That is too brittle. A single event can be noisy, data-thin, structurally unusual, or simply unrepresentative. For a multi-event model family, the statistical object should be an event-by-parameter performance matrix, not a single-event surface.

Bad current pattern

```
VectorBT broad screen
-> group candidates by model family
-> pick train_event = events[0]
-> compute surface stability on only that event
-> reject entire family if this surface fails
-> 0 candidates reach HftBacktest
```

Correct pattern

```
VectorBT broad screen
-> build event x parameter performance matrix
-> evaluate robust regions across training events / folds
```

```
-> run WFC / walk-forward / CSCV / DSR / PBO checks
-> send only robust survivors to HftBacktest
```

3. Why Restarting Is Not the Right Next Step

A full restart without changing the bridge likely repeats the same failure mode: VectorBT promotes candidates, the bridge applies the same first-event hard gate, robustness packaging accepts zero families, and HftBacktest receives nothing. That burns time and compute without changing the causal mechanism.

The correct next step is a targeted diagnostic that reuses the existing VectorBT artifacts and asks: would the same promoted candidates pass under a mathematically sane surface definition?

4. Required Bridge Sensitivity Study

Implement a diagnostic-only mode that compares the current bridge against pooled, median, and fold-level surface definitions. Do not change production HftBacktest routing until this report is generated and reviewed.

Surface Policy	Definition	Purpose
current_first_event	Current logic: use events[0] only	Baseline reproduction of current 0 / 50 result
pooled_train_events	Aggregate parameter surface over all usable training events	Main corrected low-friction methodology
median_event_surface	Compute surface per event; use median and downside dispersion	Robust to one noisy or bad event
fold_is_surface	Compute surface on in-sample events inside each WFC/walk-forward fold	Best fit if fold definitions already exist

5. Files to Touch

File	Required Change
scripts/build_robustness_raw_inputs_from_screening.py	Add --surface-policy modes and sensitivity report. Replace hard dependency on events[0] for non-baseline policies.
packages/research_pipeline/types.py	If needed, add dataclasses for surface policy diagnostics and bridge sensitivity rows. Keep additive.
packages/research_pipeline/evaluation.py	Do not rewrite event evaluation. Only consume new bridge diagnostics if required for reporting.
scripts/run_pipeline.py	Add a diagnostic-only entry point or pass-through flag only if existing orchestration owns this handoff.
docs/research/AUTORESEARCH_PIPELINE.md	Document bridge surface policies, when to use them, and why first-event gating is diagnostic only.
tests / appropriate pipeline tests	Add deterministic fixtures proving baseline reproduction and pooled/fold policy behavior.

6. Implementation Requirements

Add bridge sensitivity modes:

```
--surface-policy current_first_event
--surface-policy pooled_train_events
```

```
--surface-policy median_event_surface
--surface-policy fold_is_surface
```

Policy behavior

- **current_first_event:** Must reproduce current behavior exactly. This is required for regression comparison.
- **pooled_train_events:** Build the parameter surface using all usable training events in the family. Aggregate each parameter cell across events using median or trimmed mean. Require minimum event coverage per cell.
- **median_event_surface:** Compute the surface and stability metrics separately per event, then summarize median stability, lower-quartile stability, and event dispersion.
- **fold_is_surface:** For each walk-forward or WFC fold, compute surface stability only on the in-sample training events for that fold. Then measure whether the selected parameter region persists out-of-sample.

Coverage requirements

- Do not allow one event to reject a model family unless the usable-event coverage falls below the minimum threshold.
- Exclude events flagged as `data_bad`, `insufficient_trades`, `missing_surface`, or `nonfinite_metrics` from surface aggregation, but report them explicitly.
- Require a minimum number of parameter cells and minimum number of usable training events before evaluating surface stability.
- Separate packaging eligibility from robustness pass. Packaging should answer "is there enough evidence to test?" Robustness should answer "does it pass?"

7. Diagnostic Report Contract

Produce the following artifact:

```
runtime/robustness/robustness_bridge_sensitivity_report.json
```

Required fields:

Field	Purpose
<code>model_family</code>	Family identifier being tested
<code>vectorbt_promoted_count</code>	Candidate count entering the bridge from VectorBT
<code>event_count / usable_event_count / rejected_event_count</code>	Coverage and data-quality accounting
<code>parameter_cell_count</code>	Number of evaluated parameter cells
<code>*_pass</code> booleans	Pass/fail under current, pooled, median, and fold policies
<code>candidates_passing_*</code>	Candidate count by policy
<code>event_0_id / event_0_surface_metrics</code>	Diagnostics for current first-event behavior
<code>pooled_surface_metrics</code>	Diagnostics for corrected pooled-event behavior
<code>reason_*_failed</code>	Human-readable failure reason by policy
<code>candidates_rejected_by_current_but_passed_by_corrected_policy</code>	Most important list for determining whether the bridge is choking candidates

8. Bridge Attrition Report

Every run must explain where candidates are lost. Add or update an attrition report showing the following sequence:

Stage	Metric
VectorBT screen	promoted candidates
Family formation	model families formed
Packaging eligibility	families with enough events/cells/trades/data quality
Surface stability current	families/candidates passing current_first_event
Surface stability corrected	families/candidates passing pooled/median/fold policy
WFC / walk-forward	families/candidates passing fold persistence
CSCV / PBO	families/candidates passing overfit controls
DSR	families/candidates passing deflated Sharpe criteria
HftBacktest eligibility	candidates allowed into expensive realism replay

9. Acceptance Criteria

- The diagnostic must run against existing VectorBT screening artifacts. No full rerun is required for the first diagnostic.
- current_first_event must reproduce the current 0-family pass result, unless artifacts have changed.
- The report must show how many families and candidates would pass under each surface policy.
- If corrected policies produce nonzero survivors, the bridge is confirmed too brittle and must be fixed before HftBacktest routing.
- If all corrected policies still produce zero, then VectorBT promoted candidates were not robust enough under the available evidence.
- No candidates should be sent to HftBacktest through the old first-event gate after this issue is identified.

10. Production Recommendation After Diagnostic

Diagnostic Outcome	Recommended Action
current_first_event = 0, corrected policy > 0	Fix bridge. Use fold_is_surface if folds exist; otherwise pooled_train_events with coverage and downside-stability thresholds. Then package survivors.
all policies = 0	Do not HftBacktest these candidates. Treat VectorBT promotions as cheap-screen false positives. Improve hypotheses/features/data coverage.
current_first_event > 0 and corrected policy consistent	Current gate may not be choking this run, but still avoid single-event default for future family-level screening.

11. Developer Prompt

Use this block directly with the developer.

We need to diagnose the VectorBT -> robustness -> HftBacktest handoff before relaunching the full pipeline.

Current observation:

- VectorBT promoted 1,728 candidates.
- Robustness packaging accepted 0 / 50 model families.
- Therefore 0 candidates reached HftBacktest.
- Suspicious bridge logic is in scripts/build_robustness_raw_inputs_from_screening.py where train_event = events[0] is used.
- This means one arbitrary first event can reject an entire model family before WFC / walk-forward / CSCV / DSR / PBO evidence is constructed.

Do not restart the full pipeline yet.

Implement a diagnostic-only branch first.

Objective:

Determine whether the current first-event surface gate is choking off good VectorBT candidates.

Tasks:

1. Add bridge sensitivity modes to scripts/build_robustness_raw_inputs_from_screening.py:

```
--surface-policy current_first_event
--surface-policy pooled_train_events
--surface-policy median_event_surface
--surface-policy fold_is_surface
```

2. current_first_event must reproduce current behavior exactly.

3. pooled_train_events:

- Build the parameter surface using all usable training events in the model family.
- For each parameter cell, aggregate across events using median or trimmed mean.
- Require minimum event coverage per parameter cell.
- Do not let one event reject the entire family unless coverage falls below threshold.

4. median_event_surface:

- Compute the parameter surface separately for each event.
- Compute stability metrics per event.
- Use median stability and downside/worst-quartile stability across events.
- Report event dispersion.

5. fold_is_surface:

- For each walk-forward fold, compute surface stability only on the in-sample training events for that fold.
- Record whether the selected parameter region persists out-of-sample.
- This is the preferred methodological mode if fold definitions already exist.

6. Add robustness_bridge_sensitivity_report.json with per-family pass/fail counts, reasons, event_0 diagnostics, pooled diagnostics, and candidates rejected by current but accepted by corrected policy.

7. Do not change HftBacktest routing yet. This is diagnostic only.

8. Acceptance criteria:

- Running the diagnostic on existing VectorBT artifacts must show how many families/candidates pass under each surface policy.
- If current_first_event gives 0 but pooled/fold gives nonzero, the bridge is confirmed too brittle.
- If all policies give zero, VectorBT promoted candidates were not robust and the pipeline correctly stopped before HftBacktest.

9. After diagnostic, propose default production policy. Expected default: fold_is_surface if fold definitions are available; otherwise pooled_train_events with event-coverage and downside-stability thresholds.

Hard constraints:

- Do not treat a single event surface as a family-level prerequisite.
- Do not send candidates to HftBacktest until the bridge is validated.
- Do not relaunch the full pipeline until this diagnostic proves whether the bridge is choking candidates or candidates are genuinely non-robust.

12. Plain-English Decision Tree

```
Do not restart full pipeline yet.
|
v
Run bridge sensitivity diagnostic from existing VectorBT artifacts.
|
+-- corrected policies still produce 0 survivors
| -> models/candidates are probably not robust enough
| -> improve hypotheses/features/data; do not HftBacktest
|
+-- corrected policies produce nonzero survivors
-> current bridge is choking candidates
-> fix bridge policy
-> then send corrected survivors to HftBacktest realism
```

13. Notes on Academic Rationale

The key statistical principle is that model selection over many candidates needs an event-by-parameter or observation-by-trial matrix. A single event is not enough to infer family-level parameter surface stability. Robustness should be established by repeated train/test splits, walk-forward evidence, multiple-testing adjustment, probability of backtest overfitting, and deflated performance metrics.

In practice, the bridge should build the evidence object first, then let robustness methods reject or accept. It should not reject the family before the evidence object exists.

14. Final Recommendation

The consensus is correct: VectorBT did its broad-screen job. The suspected failure is the robustness handoff. The next move is a bridge sensitivity diagnostic, not a full rerun. If corrected surface policies produce survivors, fix the bridge. If they do not, the models likely do not have robust alpha under current features/data and should not be sent to HftBacktest.

References / Methodology Anchors

This document is based on the reported run outcome, the identified first-event surface gate, and standard quantitative validation concepts: event-by-parameter performance matrices, walk-forward validation, combinatorially symmetric cross-validation, probability of backtest overfitting, deflated Sharpe ratio, and multiple-testing control. It is an implementation prompt, not a guarantee of alpha.